

From Production to Perception

Designing a Sociophonetic Study

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Today's Roadmap



Production Background

/s/ variation, prosodic patterns, and persona guises



Why Perception?

Theoretical motivation and the L1/L2 question



Speaker Selection

Acoustic analysis and design decisions



Matched-Guise Design

Paired comparison and two-pass structure



Implementation

Qualtrics, counterbalancing, measures, and analysis



Challenges & Lessons

Stimulus prep, fatigue, operationalizing identity

The Production Study

Kelley (2024) | Third-wave variationism: speakers actively use sound variation to construct social identity (Eckert, 2008, 2012)

13 speakers from Seville, Spain performed *gay* and *straight* persona guises (adopting characteristics associated with each identity) while reading controlled passages. Acoustic analysis revealed systematic variation in two domains:

Sibilant variation

Two properties of /s/ sounds shifted: duration (how long /s/ lasts) and center of gravity or COG (the dominant frequency — how sharp or bright the sound is). Gay guises showed longer, higher-COG /s/ (Munson et al., 2006; Mack, 2010).

Prosodic differentiation

Speakers also modulated prosodic features (pitch range, intonation contours, and speech rate) as part of their guise performances (Kelley, in prep).

The perception study focuses on /s/ variation because /s/ sounds are discrete, individual segments that can be precisely measured and compared one by one. Prosodic cues (rhythm, pitch, and intonation) are spread across the entire utterance and difficult to isolate.

Why Perception?

Three motivations for moving from production to perception:

The production gap

We know speakers change their /s/ sounds across persona guises, but a measurable acoustic difference does not guarantee that listeners actually notice it (Drager, 2011). Listeners may not attend to the same features researchers measure.

Social meaning

Even if listeners detect the contrast, do they assign social meaning to it? Do they hear one guise as "gayer" or "more masculine"? Production data cannot answer this (Campbell-Kibler, 2009; Walker et al., 2014).

The L1/L2 question

Native (L1) and non-native (L2) speakers hear the same sound signal but bring different social knowledge. L2 learners may hear the acoustic difference but not know what it socially means, because they have not internalized the same associations between sounds and social categories that L1 speakers have (Chappell & Kanwit, 2022).

Research Questions

RQ1

When listeners hear two recordings of the same speaker (one gay guise, one straight guise), can they tell which is which?

RQ2

Do native Spanish speakers (L1) and second-language learners (L2) differ in their ability to identify guises?

RQ3

When the acoustic difference between guises is larger (in /s/ duration or COG), are listeners more accurate?

RQ4

Do listener judgments of personality traits (e.g., masculinity, confidence) shift after they learn the study is about sexual orientation (SO)?

Expanding Across Dialects

Why three dialect regions?

Sevillian

Syllable-final /s/ weakens
(aspiration, deletion)

2 speakers

Venezuelan

Variable /s/ at end of
syllables

2 speakers

Basque

Apicoalveolar /ɕ/
Full /s/ retention

1 speaker

*Each region produces /s/ differently: testing whether listeners' perception generalizes across these different sound systems
(Walker et al., 2014; Boyd et al., 2021; Hualde, 2005)*

Speaker Selection: From 7 to 5

Decision framework: How do you choose which speakers to include?

Acoustic contrast

Both duration and COG should differentiate in the expected direction (Munson et al., 2006; Mack & Munson, 2012)

Effect size range

Include speakers with large differences AND one with almost no difference (a control) to test whether perception tracks the size of the acoustic gap

Design balance

Represent all three dialects, both speaker SO groups, both passage types

Eliminate confounds

No overlapping passage text; no reversed acoustic patterns

Eliminated: Speaker H5 (Sevillian): /s/ frequency was higher in the straight guise (opposite of expected). Speaker L8 (Venezuelan): same passage text as another speaker, creating a familiarity confound.

Stimulus Acoustic Properties

Speaker	Dialect	SO	Pass.	n	Gay dur (ms)	Str dur (ms)	d(dur)	Gay COG (Hz)	Str COG (Hz)	d(COG)
Sp1 (L9)	Venezuelan	LGBTQ+	Narr.	16/16	111.2	78.1	0.66	4751	2954	0.90
Sp2 (L7)	Venezuelan	LGBTQ+	Sci.	19/19	76.3	74.3	0.04	2629	2431	0.09
Sp3 (H9)	Basque	non-LGBTQ+	Narr.	16/17	65.6	63.5	0.08	5599	3858	1.41
Sp4 (H8)	Sevillian	non-LGBTQ+	Sci.	20/20	87.5	69.0	0.55	5806	4270	0.95
Sp5 (L2)	Sevillian	LGBTQ+	Sci.	7/7	170.0	85.7	2.69	6099	4973	1.89

Note. d = Cohen's d, a standardized measure of difference (0.2 = small, 0.5 = medium, 0.8 = large). Green bold = statistically significant. Gray = near-null control pair. COG = center of gravity (dominant frequency of the /s/ sound). dur = duration in milliseconds.

Discussion point: Speaker 3 (H9, Basque) shows a very large frequency difference ($d = 1.41$) but almost no duration difference. The guise contrast comes entirely from the quality of the /s/ sound, not how long it lasts. This gives us a natural test of whether listeners respond to one acoustic property independently of the other.

Design Balance

Dialect

Venezuelan (2), Sevillian (2),
Basque (1)

Speaker SO

LGBTQ+ (3), non-LGBTQ+ (2)

Passage

Narrative (2), Science (3)

Acoustic Contrast Gradient

Sp2 (L7)

Null control

d = 0.09

Sp3 (H9)

COG only

d = 1.41

Sp1 (L9)

Strong

d = 0.90

Sp4 (H8)

Strong

d = 0.95

Sp5 (L2)

Maximum

d = 1.89

Low contrast

High contrast

Excerpt Selection: Competing Constraints

For each speaker, select a ~10–14 second continuous excerpt that:

1. Maximizes density of syllable-final /s/ tokens
2. All (or nearly all) /s/ tokens differentiate in the expected direction
3. Forms a self-contained narrative or informational unit
4. Avoids textual overlap with other speakers' excerpts

Case study: Speaker 3 (H9, Basque)

Candidate A (Mirror dialogue): 20 /s/ tokens, but 4 with reversed effects

Candidate B (Cazador command): 10 tokens, all positive, but only 7.7 seconds

Selected: Mirror dialogue + cazador combined: 16 tokens, 100% in expected direction, $\Delta\text{COG} +1576$ Hz, ~13 seconds

The Matched-Guise Paradigm

Comparing two recordings from the same speaker controls for individual voice differences (Lambert et al., 1960; Walker et al., 2014).

What participants see

♪ Grabación A

♪ Grabación B

"Which speaker seems more ____?"

What's actually happening

Same speaker, same passage

A = gay guise B = straight guise

(or reversed for BA group)

Visual Analogue Scale (VAS): a continuous 0–100
slider per trait

Neg. anchor ——— Pos. anchor

Counterbalancing: Half of participants get AB order (gay = A), half get BA (gay = B). Pair order randomized within each participant.

The Two-Pass Design

Separating judgments made before vs. after participants know the study involves sexual orientation

Pass 1: General Traits

All 5 pairs, no mention of SO

Bipolar VAS trait ratings (0–100):

- Pleasantness
- Confidence, Education
- Masculinity, Familiarity

per speaker (Yang & Jeon, 2021)



Pass 2: SO + Trait Ratings

Same 5 pairs, after disclosure

- SO identification (forced-choice)
- Confidence VAS + re-rate:

Same 5 VAS dimensions

(priming shift = Pass 2 – Pass 1)

The transition screen: "The following questions will ask for more specific perceptions about the speakers' personal characteristics." Participants can withdraw at this point. This serves both ethical (informed consent) and methodological (priming control) purposes.

Why Two Passes? Decision Rationale

Alternative: Single consolidated loop

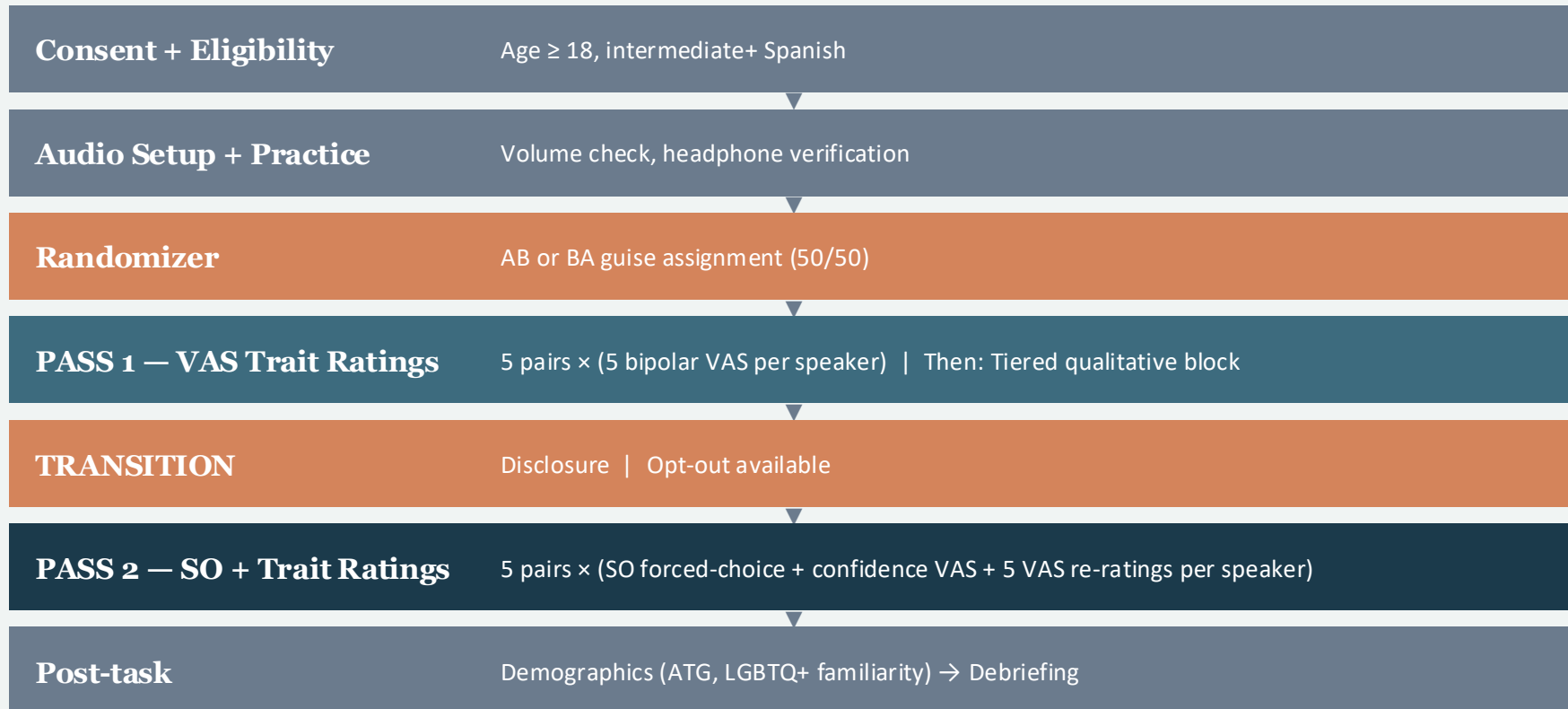
- + Simpler Qualtrics implementation
- + Participants hear each pair only once
- + Shorter overall duration
- SO question appears with no warning
- All traits collected post-priming
- Cannot compare primed vs. unprimed

Chosen: Two-pass with transition

- + Unprimed baseline trait data (Pass 1)
- + Gradual disclosure = ethical safeguard
- + Can test priming effect as a DV
- Participants re-hear all pairs
- Longer session (~5–10 min added)
- Complex Qualtrics architecture

Discussion: *Is the added participant burden justified by the ability to measure priming effects? What would you prioritize?*

Survey Flow Architecture



Target total: ~30 minutes (VAS format reduces rating time vs. Likert scales)

Outcome Measures & Covariates

Dependent Variables

Primary: SO identification (forced-choice, recoded for accuracy)

Secondary:

- 5 bipolar VAS trait ratings (0–100) per speaker
- Tiered metalinguistic awareness (4-tier protocol)
- Confidence VAS (0–100)
- VAS shift (Pass 2 – Pass 1 per dimension)

Covariates

Speaker-level:

- Pre-measured d(dur) and d(COG)
- Dialect region, passage type

Listener-level:

- L1 vs. L2 group
- ATG score (attitudes toward gay men)
- LGBTQ+ familiarity, dialect familiarity
- L2 proficiency, immersion experience

Planned analysis: Statistical models that account for the fact that each listener rates multiple speakers and each speaker is rated by multiple listeners. Predictors include listener group (L1/L2), dialect, and acoustic contrast size. The shift in ratings between Pass 1 and Pass 2 is analyzed as a direct measure of the priming effect.

Tiered Metalinguistic Awareness Block

A step-by-step approach that measures what speech features listeners notice on their own, without ever directly asking about /s/ (Preston, 1999; Schmidt, 2022; Chappell & Kanwit, 2022)

Tier 1: Free Description (all participants)

Open text: "¿Cómo describiría las diferencias entre las voces?" Code for spontaneous /s/ mention.

Tier 2: Feature Checklist (all participants) — select which features varied: vowels, consonants, speed, intonation, accent, rhythm, clarity, voice quality. Target ("consonantes") embedded among distractors. Randomized order.

Tier 3: Conditional Elaboration (display logic)

Only appears if participant selected "consonantes" in Tier 2. Open text: elaborate on consonant differences. Non-selection is itself data.

***Tier 4: Dialect Placement (all participants)** Where is each speaker from? Dropdown with 8 regional options. An indirect measure of /s/ sensitivity: identifying Spanish dialects depends heavily on how /s/ is pronounced. Key question: do L2 learners notice consonant differences (Tier 2) without being able to place the speaker in the correct region (Tier 4)?*

Challenges & Lessons Learned

Stimulus preparation

Forced alignment (automatic matching of audio to transcription), acoustic measurement extraction, and excerpt selection: each step introduces decisions that shape what participants ultimately hear. Even automated tools require manual verification at every stage.

Participant fatigue

5 pairs × 2 passes × trait ratings × demographics pushes the session close to 35 minutes. Every added question has to justify its analytical payoff against the cost of participant attrition and declining attention.

Operationalizing identity

Asking "which speaker sounds more likely to be gay?" treats sexual orientation as a binary perceivable from speech — a simplification the study requires but that the debriefing must address (Butler, 1990; Eckert, 2008; Podesva, 2007).

Production-to-perception translation

A large measured acoustic difference does not guarantee listeners can hear it. The size of the difference in production does not straightforwardly predict whether listeners perceive it. It may fall below the threshold of what humans can detect.

Open Questions for Discussion

- 1 Is the two-pass structure worth the added participant burden?
- 2 Should trait ratings be collected for both speakers or just one per pair?
- 3 How should we handle the /s/-specific qualitative question?
- 4 What's the right sample size for detecting a priming effect?
- 5 Are there perception methodology precedents for this design in Spanish sociolinguistics?

Selected References

/s/ & perceived sexual orientation

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L1/L2 & cross-linguistic perception

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Thank You

Questions, feedback, and discussion welcome.

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